

Biochemistry Practical

Principles, exact course protocols, interpretation, calculations and laboratory reasoning

Updated 8 September 2026

All seven organized laboratory items are reviewed, including the two earlier foundation PDFs linked into this folder. The two qualitative-test versions are reconciled without concealing protocol differences. Equipment and assay explanations are for study and supervised university practicals, not unsupervised chemical experimentation or patient treatment. No practical recordings or specimen-specific flame-photometer SOP were found.

How to use this book

Lecture-grounded material and book extensions are distinguished. Green/blue panels consolidate the key ideas; amber panels flag missing evidence; orange panels correct traps. This is a detailed review of the available sources, not an official exam blueprint. Original labeled figures remain important. Numerical laboratory ranges and procedures depend on the course method and local SOP.

Original lecture transcripts are preserved as PDF attachments and in the editable source folder when available. Automated transcripts can contain errors; study the checked synthesis first. Personal study only: original book/lecture illustrations retain their owners' rights.

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Practical map: what you must be able to do

SOURCE BASIS: lecture

Learn every experiment in the same order: analytical question, reaction or measurement principle, equipment, exact course mixture, independent variable and controls, expected observation, interpretation, calculation and sources of error. A positive colored tube is evidence of a chemical property, not automatic identification of one unique molecule. A numerical result is meaningful only after checking the units, blank, calibration and dilution.

Station	Core task	Reasoning beyond memorization
Safety and equipment	Identify apparatus and select an appropriate measuring device	Separate approximate holding vessels from calibrated volumetric devices; explain safe handling and waste routes.
Titration	Determine acid concentration using standardized NaOH	Use delivered buret volume, normality and reaction equivalence; distinguish an indicator endpoint from equivalence.
Spectrophotometry	Select wavelength; prepare standards; assay total protein	Relate absorbance to concentration and transmittance; know what a blank removes and what it cannot remove.
Qualitative protein/amino-acid tests	Recognize reactive groups and course color reactions	Discriminate overlapping positive tests, limitations, control tubes and version-specific timings.
Enzymes	Interpret milk-clotting and urease experiments	Separate enzyme concentration, substrate concentration, temperature, specificity and inhibition; do not confuse clot time with a calibrated initial velocity.
Flame photometry	Explain emission, components and electrolyte calibration	Separate emission from absorption, analytical wavelength from perceived color, and true electrolyte change from sample or method artifact.

The answer structure for an oral station

State the principle in one sentence, then the exact observation and its chemical explanation. Only then calculate or identify. For an abnormal result, offer one biological explanation and one pre-analytical/analytical alternative.

Sources for this section

- 01 - Laboratory safety and equipment.pdf - PDF pp. 1-46
- 02 - Titration practical and calculations.pdf - PDF pp. 1-10
- 03 - Spectrophotometry and Beer-Lambert law.pptx - entire deck
- 04A - Amino acid and protein qualitative tests - Ebrahimi.pptx - entire deck
- 04B - Amino acid and protein qualitative tests - Emamgholipour.pdf - PDF pp. 1-28
- 05 - Enzyme practical.pptx - entire deck
- 06 - Flame photometry and body electrolytes.pptx - entire deck

Safety, waste and apparatus handling

SOURCE BASIS: lecture

Wear the laboratory coat, appropriate gloves and eye protection; tie back hair and know the eyewash, shower, exits and fire equipment. Do not eat, drink or mouth-pipette. Read labels before dispensing, avoid returning excess reagent to stock bottles, and keep samples clearly identified. Gloves do not replace hand washing or make it acceptable to touch personal devices with contaminated hands.

Health Hazardous Chemicals

Dangerous chemical signs

Health Hazard 	Flammability 	Compressed Gas 
Corrosive 	Explosive 	Oxidizers 
Environmental 	Acute Toxicity 	Other Hazards 

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Recognize the hazard pictograms shown in the course equipment PDF. Identify the chemical hazard before selecting handling and disposal precautions.

Identify: Flammable; Corrosive; Oxidizer; Compressed gas; Acute toxicity; Health hazard; Environmental hazard

Figure source: Laboratory safety and equipment, PDF p. 10

- When a protocol requires acid dilution, add acid slowly to water, with suitable heat control; do not pour water into concentrated acid. Heat of dilution can cause violent local boiling and splashing.
- The qualitative tests use concentrated nitric or sulfuric acid, strong NaOH, TCA and acetone-containing ninhydrin. They require supervision, suitable ventilation and the designated chemical-waste route. Ninhydrin in acetone is flammable: a boiling-water bath is not permission to use an open flame beside it.
- Do not improvise spill neutralization or mix waste streams. Follow the course/lab procedure and alert the supervisor. An ordinary fume hood removes hazardous vapors; it is not interchangeable with a biological safety cabinet.
- Balance centrifuge tubes opposite one another by appropriate mass and compatible containers; secure the rotor/lid and wait for complete stopping. The centrifuge separates materials according to sedimentation behavior, not by magically identifying a molecule.
- A hot container may look cold. Use the correct holder, point heated tubes away from people, and do not seal a vessel being heated unless it is specifically designed for that procedure.

Sources for this section

01 - Laboratory safety and equipment.pdf - PDF pp. 5-12, 39-44

Choose the correct measuring device

SOURCE BASIS: lecture

Device	Use	Important limitation
Beaker / Erlenmeyer flask	Hold, mix and, for Erlenmeyer, swirl a titration mixture	Graduations are approximate; neither is the correct tool for an exact analytical aliquot.
Florence flask	Uniform heating of liquid	Not a volumetric standard merely because it is a flask.
Volumetric flask	Prepare one accurately defined final volume	Dissolve first and bring the meniscus to the calibration mark; do not add the full stated volume of solvent to an existing solute volume.
Graduated cylinder	Measure routine liquid volumes	Less precise than a suitable volumetric pipette for a fixed analytical aliquot.
Volumetric pipette	Transfer one calibrated volume	Use a filler and follow its calibration/drainage markings, rather than automatically blowing out every pipette.
Mohr graduated pipette	Deliver variable graduated volumes	Graduations stop before the tip; measure between appropriate marks.
Serological pipette	Deliver graduated volumes extending toward the tip	Whether to blow out follows the pipette marking/type; recognize it separately from Mohr.
Buret	Deliver a measured variable volume through a stopcock	Delivered volume equals final reading minus initial reading; exclude bubbles and read at eye level.
Disposable transfer pipette	Transfer drops or approximate volumes	Do not treat an uncalibrated plastic dropper as a precise quantitative pipette.
Balance, spatula, watch glass	Weigh solid using a suitable vessel	Tare correctly, avoid contamination and allow stable readings.
Funnel and filter paper	Transfer liquid or separate suspended solid	Filtration does not remove dissolved analyte.
Glass rod / magnetic stir bar	Mix solutions	Avoid losses and contamination; stirring is not a substitute for complete dissolution.

MOHR AND SEROLOGICAL PIPETTES

Measuring pipettes are divided into:

- **MOHR PIPETTES**
– the graduations on these always end before the tip
- **SEROLOGICAL PIPETTES**
– the graduation marks continue to the tip



19

The course compares graduated pipette types. Track where the graduations terminate before deciding how a measured volume is delivered.

Identify: Mohr pipette; Serological pipette; Tip; Graduations

Figure source: Laboratory safety and equipment, PDF p. 19

For a clear aqueous solution with a concave meniscus, read the bottom at eye level; viewing above or below introduces parallax. Use the specified reading convention for an opaque or differently wetting liquid. Temperature affects volumetric calibration. A water bath controls temperature of an immersed sample; a hotplate supplies heat; an oven is not automatically a validated sterilization process for every material.

Check yourself: Why does adding 10 mL of water to a solute not necessarily make a 10 mL solution?

Volumes need not be additive. A volumetric preparation dissolves the solute and then adjusts the total solution volume to 10 mL. Concentration uses final solution volume, not the volume of solvent initially added.

Sources for this section

01 - Laboratory safety and equipment.pdf - PDF pp. 13-38

Titration: neutralization, equivalence and endpoint

SOURCE BASIS: lecture

The course titrates an unknown acid against standardized 0.1 N sodium hydroxide using phenolphthalein. Normality means reactive equivalents per liter for the specified reaction. At stoichiometric equivalence, acid equivalents equal base equivalents, so $N_{\text{acid}} \times V_{\text{acid}} = N_{\text{base}} \times V_{\text{base}}$ when volumes use the same units. Molarity equals normality only when one equivalent reacts per mole; for a polyprotic acid it depends on which neutralization stage is being measured.

- Phenolphthalein is colorless in acid and pink in the alkaline transition range. The course endpoint is a pale pink color persisting at least 30 seconds, not an intense pink overshoot.

- The endpoint is the observed indicator change; the equivalence point is the stoichiometric completion. They are designed to be close but are not identical concepts. Equivalence pH is not universally 7, especially for weak-acid or weak-base titrations.
- Record actual initial and final buret readings. The slide formula using final reading as volume is valid only if the initial reading is zero. Burets commonly have increasing readings downward: a higher final reading indicates delivered liquid.
- The course says about 10 mL acid for setup. A quantitative concentration calculation needs the accurately measured actual aliquot, not an unverified approximate beaker volume.

Step	Course procedure	Why it matters
Prepare	Measure the acid aliquot and add 2-3 drops of phenolphthalein	Excess indicator can add unnecessary chemical/visual interference.
Buret	Fill with standardized 0.1 N NaOH, remove tip bubbles, record initial reading	Volume filling a previously empty tip is not necessarily delivered to the acid.
Titrate	Swirl and add NaOH; slow to dropwise near the color change	Rapid addition near endpoint makes overshoot likely.
Endpoint	Pale pink remains for at least 30 seconds	A transient local pink swirl does not establish completion.
Record	Read the final buret value to the course-specified 0.01 mL estimation	Use final minus initial, retain appropriate significant figures and units.
Calculate	Use $N_{acid} = N_{base} \times (V_{final} - V_{initial}) / V_{acid}$	Do not silently replace normality by molarity.

Sources for this section

02 - Titration practical and calculations.pdf - PDF pp. 1-10

Titration calculations and directional errors

SOURCE BASIS: lecture

Worked example, not a measured class result: 10.00 mL acid requires a buret change from 0.60 to 12.80 mL of 0.100 N NaOH. Delivered NaOH is 12.20 mL, giving acid normality $0.100 \times 12.20 / 10.00 = 0.122$ N. If a particular acid supplies two equivalents per mole at this endpoint, its molarity would be 0.0610 mol/L; that factor cannot be assumed before the acid and reaction are known.

Problem	Effect on reasoning/result	Prevention
Over-titration	Too much base recorded; calculated acid concentration is too high	Approach endpoint dropwise while mixing.
Tip bubble fills during titration	Buret change can exceed volume reaching flask; calculated acid may be too high	Fill the tip and remove bubbles before initial reading.

Problem	Effect on reasoning/result	Prevention
Final reading used instead of delivered volume	Positive initial reading is mistakenly included; concentration too high	Subtract initial reading.
Buret not conditioned and base diluted by residual water	Nominal base concentration exceeds actual concentration; using nominal value can overestimate acid	Condition with titrant as instructed and use standardized concentration.
Acid aliquot measured inaccurately	Wrong denominator changes calculated concentration	Use the correct volumetric device and actual aliquot.
Water added to acid flask after measuring acid	Does not change the acid equivalents, provided no sample is lost	Distinguish diluting the titration mixture from changing the original aliquot.
Unmixed local indicator color	False early endpoint may underestimate acid	Swirl thoroughly and apply persistent-color criterion.

Check yourself: Does a 10.00 mL acid aliquot need more NaOH after adding 20 mL distilled water to its flask?

No, not for neutralization of the same acid amount if the water introduces no acid/base and there is no sample loss. Concentration inside the flask is lower, but the number of acid equivalents is unchanged.

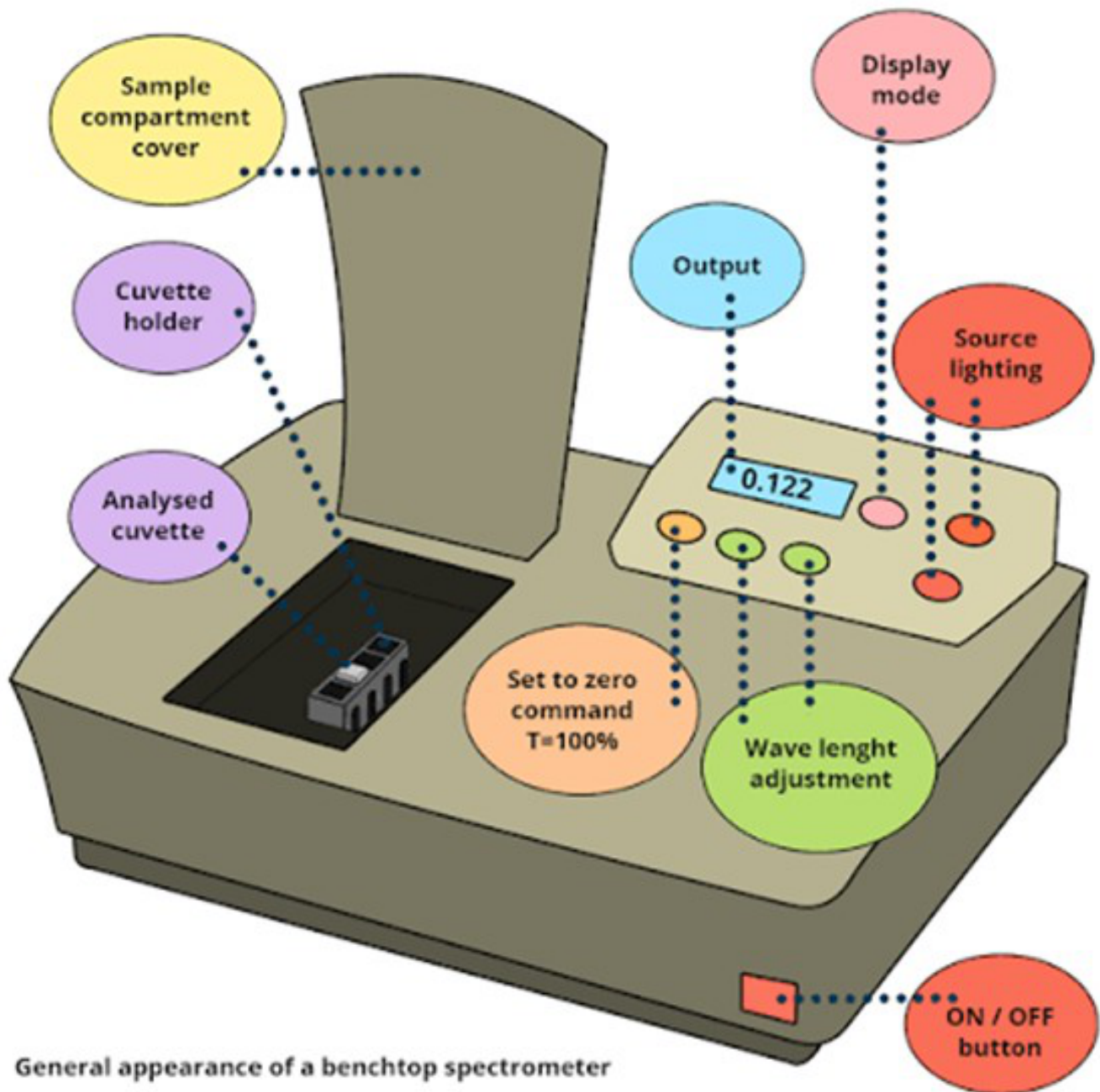
Sources for this section

02 - Titration practical and calculations.pdf - PDF pp. 8-10

Spectrophotometry: light path, wavelength and cuvette

SOURCE BASIS: lecture

A spectrophotometer measures how much light a sample transmits at a selected wavelength. Its core sequence is light source, wavelength selection, sample cuvette, detector and signal/readout. A collimator directs the beam; a monochromator or suitable filter isolates a spectral region. The course uses tungsten illumination for visible work and deuterium for ultraviolet work. UV analysis requires a cuvette material transmitting the selected UV wavelength, usually suitable quartz; ordinary glass and many plastics absorb UV.



General appearance of a benchtop spectrometer

Course instrument diagram: locate the wavelength selector, sample compartment, blank/zero control and absorbance display before operating an unfamiliar unit. The exact buttons depend on the laboratory instrument.

Identify: Wavelength adjustment; Cuvette holder; Analyzed cuvette; Display; Blank/zero control

Figure source: Spectrophotometry and Beer-Lambert law, slide 6, embedded source diagram

The wavelength absorbed is not necessarily the color seen: a colored solution preferentially removes some wavelengths and transmits their complementary mixture. Select the absorption maximum for a sensitive assay when the protocol permits. The course gives approximate visible and UV intervals; these are conventions, not sharp physiological boundaries. Keep the same optical path length, orient cuvettes consistently, avoid fingerprints on optical faces, remove bubbles and ensure enough liquid covers the beam.

Transparent is not always optically suitable

A cuvette may look clear to your eye yet absorb ultraviolet light. Similarly, a cloudy sample scatters light: its apparent absorbance is not automatically due to the desired chromophore.

Sources for this section

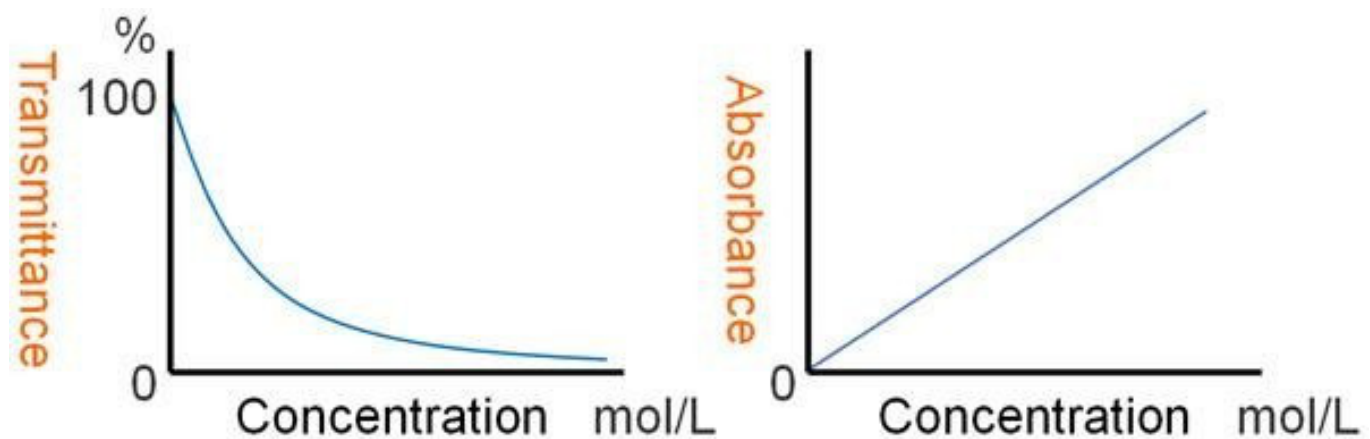
03 - Spectrophotometry and Beer-Lambert law.pptx - slides 1-11

Beer-Lambert law, absorbance and transmittance

SOURCE BASIS: lecture

Transmittance $T = I/I_0$, where I_0 is the incident/blank-reference intensity and I is transmitted intensity. Absorbance $A = \log_{10}(I_0/I) = -\log_{10}(T)$. If transmittance is expressed as a percentage, $A = 2 - \log_{10}(\%T)$. Absorbance is dimensionless. The difference I_0 minus I is absorbed intensity, not optical density; slide 13 incorrectly equates that difference with absorbance.

Beer-Lambert law is $A = \epsilon \times b \times c$. With c in mol/L and path b in cm, molar absorptivity ϵ has units $L \text{ mol}^{-1} \text{ cm}^{-1}$. If mass concentration is used, the corresponding specific absorptivity must use matching mass units. At a fixed wavelength, path length and chemical state, absorbance is proportional to concentration within the validated range. A doubling of concentration doubles A but does not halve transmittance in general.



The source curves contrast exponential loss of transmittance with linear increase of absorbance as concentration rises. This is why calibration ordinarily uses absorbance rather than percent transmittance.

Identify: Percent transmittance; Absorbance; Concentration

Figure source: Spectrophotometry and Beer-Lambert law, slide 14, embedded graphs

%T	T	A
100	1.00	0
50	0.50	about 0.301
10	0.10	1
1	0.01	2

Check yourself: A solution has $A = 0.50$ in a 1 cm cuvette. What happens in a 2 cm path at the same concentration, if Beer-Lambert behavior holds?

Absorbance doubles to 1.00. Transmittance changes from about 31.6% to 10%, not to 15.8%. Path length multiplies absorbance because absorbance is logarithmic in transmitted intensity.

Sources for this section

03 - Spectrophotometry and Beer-Lambert law.pptx - slides 12-16

Blanks, calibration, dilution and optical errors

SOURCE BASIS: lecture

A reagent blank contains the relevant solvent and reagents but no target analyte. It corrects their baseline optical contribution and the measurement system under those conditions. A water blank is appropriate only when it matches the specific protocol; it cannot remove all color created by unmatched reagents. A calibration standard has known analyte concentration. A control is a separate sample used to assess performance and should not be confused with the standard that defines the calibration.

- A multi-point calibration plots blank-corrected A against known concentration. Read an unknown inside the established linear range; dilute and remeasure an overly concentrated sample rather than extrapolating far beyond the standards.
- A single-standard ratio $C_{\text{unknown}} = A_{\text{unknown}}/A_{\text{standard}} \times C_{\text{standard}}$ assumes the same reaction conditions and an adequately linear, blank-corrected relationship passing through the appropriate baseline. If a fitted calibration has a meaningful intercept, use its equation instead.
- For a pre-diluted unknown, multiply the measured diluted concentration by the dilution factor: final volume/aliquot volume. Sequential dilution factors multiply. Do not add a second dilution correction merely because both standards and unknowns received the same assay reagent volume.
- High concentration, stray light, wrong wavelength, turbidity, dirty/scratched cuvettes, bubbles, inconsistent path length, incomplete color development and variable pH/temperature can break the simple relationship. A blank cannot rescue a nonlinear reaction or a mislabeled sample.
- Pipetting uncertainty and unit mistakes may dominate the result even when the spectrophotometer display gives several decimal places. Retain useful precision but do not mistake display resolution for accuracy.

Check yourself: An unknown diluted 1 mL to a final 5 mL gives 2 mg/L from the calibration. What is the original concentration?

The dilution factor is 5, so the original is 10 mg/L. If the diluted unknown and standards each later receive the same reaction mixture, that shared assay dilution is already accounted for by calibration.

Sources for this section

03 - Spectrophotometry and Beer-Lambert law.pptx - slides 15-16, 22-29

Course experiment: bromophenol-blue wavelength and standard curve

SOURCE BASIS: lecture

The course first measures a 10 mg/L bromophenol-blue solution against distilled water from 480 to 620 nm at 20 nm intervals, recording absorbance at each wavelength and plotting A against wavelength. The purpose is to identify the observed absorption maximum under these solution conditions. The phrase about increments of 480 on slide 20 is a typo; the listed wavelength sequence and slide 18 specify 20 nm increments.

Measurement order	Wavelength (nm)	Record
1	480	Measured A
2	500	Measured A
3	520	Measured A
4	540	Measured A
5	560	Measured A
6	580	Measured A
7	600	Measured A
8	620	Measured A

For the course concentration series, stock concentration is 10 mg/L and each tube has total volume 10 mL. Use $C_2 = C_1 \times V_1/V_2$. The six mixtures below therefore produce the stated standards. These are derived concentrations, not concentrations of the stock reagent before dilution.

Tube	10 mg/L stock (mL)	Water (mL)	Final concentration (mg/L)
Blank	0	10	0
1	1	9	1
2	2	8	2
3	4	6	4
4	6	4	6
5	8	2	8

Check yourself: Why measure standards at one selected wavelength rather than measuring each at a different maximum?

The absorptivity must remain comparable. Changing wavelength between standards changes the analytical response coefficient, so an absorbance difference no longer represents concentration alone.

Sources for this section

03 - Spectrophotometry and Beer-Lambert law.pptx - slides 17-24

Course total-protein assay: the quantitative Biuret method

SOURCE BASIS: lecture

In strongly alkaline solution, Cu^{2+} forms a violet complex with peptide-bond-associated nitrogen groups. The Biuret reaction detects the peptide backbone; at least two peptide bonds are needed for the classic positive reaction. Protein standards permit total-protein quantification. This is chemically related to, but procedurally distinct from, the qualitative $\text{NaOH}/\text{CuSO}_4$ tube test in the other deck.

Tube	Sample component	Biuret reagent	Incubation/readout
Blank	0.5 mL normal saline	2.0 mL	Room temperature 15 minutes; 540 nm
Standard	0.5 mL protein standard, 7 g/dL	2.0 mL	Same timing and wavelength
Unknown	0.5 mL sample	2.0 mL	Same timing and wavelength

After blank correction, $C_{\text{unknown}} = (A_{\text{unknown}}/A_{\text{standard}}) \times 7 \text{ g/dL}$, using the course standard. Example: $A_{\text{unknown}} 0.42$ and $A_{\text{standard}} 0.49$ give 6.0 g/dL. This is not multiplied by five simply because 0.5 mL sample was mixed with 2 mL reagent: the standard underwent the same assay dilution. An additional pre-dilution of the unknown would require its own factor.

Concentration is not the same as protein mass

The course lists total protein about 6-8.3 g/dL and albumin about 3.4-5.4 g/dL; reference intervals are method/lab dependent. Dehydration can raise concentration without increasing total body protein. Inflammation or a monoclonal protein can raise globulins. Blood loss, liver synthetic failure, malnutrition or protein loss may lower protein. A single concentration cannot establish which mechanism is present.

Check yourself: Why does a blue copper-containing blank not count as a positive protein result?

Blue is a property of copper reagent itself. The characteristic positive change is the peptide-associated violet complex compared with the properly matched blank, not merely the presence of any visible color.

Sources for this section

03 - Spectrophotometry and Beer-Lambert law.pptx - slides 25-31

04A - Amino acid and protein qualitative tests - Ebrahimi.pptx - slides 26-27

04B - Amino acid and protein qualitative tests - Emamgholipour.pdf - PDF pp. 24-25

Amino-acid and protein chemistry behind the qualitative tests

SOURCE BASIS: lecture + book

An alpha-amino acid has an alpha carbon bearing the amino group, carboxyl group, hydrogen and characteristic side chain. Glycine is achiral; proline constrains backbone geometry and has a secondary alpha amino group. In water, charge depends on pH and pKa. A zwitterion has both positive and negative groups; a net charge of zero does not mean that no groups are ionized. Peptide-bond formation links an alpha carboxyl group to an amino group; folding then determines the protein environment of reactive side chains.

Teaching group	Members: three-letter / one-letter codes	Test-relevant point
Nonpolar/aliphatic group	Gly G, Ala A, Val V, Leu L, Ile I, Met M, Pro P	Proline differs in ninhydrin color; methionine contains sulfur but is not cysteine.
Aromatic	Phe F, Tyr Y, Trp W	Tyr has a phenolic ring; Trp has an indole. Aromaticity alone does not make all qualitative tests equivalent.
Polar uncharged group	Ser S, Thr T, Cys C, Asn N, Gln Q	Cysteine has a thiol; Asn/Gln are amides, not the acidic Asp/Glu carboxylates.
Acidic side chains	Asp D, Glu E	Side-chain carboxyl groups are ordinarily deprotonated near neutral pH.
Basic side chains	Lys K, Arg R, His H	Histidine has an imidazole ring and is Pauli-positive; its charge varies sensitively around its relevant pKa.

These categories are teaching groupings, not mutually exclusive physical absolutes: tyrosine is aromatic and polar, cysteine and tyrosine can ionize, and a residue pKa can shift inside a protein. Protein denaturation disrupts higher-order structure without automatically hydrolyzing every peptide bond. Therefore a denatured protein can remain Biuret-positive while losing biological enzyme activity.

Sources for this section

04A - Amino acid and protein qualitative tests - Ebrahimi.pptx - slides 1-29

04B - Amino acid and protein qualitative tests - Emamgholipour.pdf - PDF pp. 1-28

Lippincott Illustrated Reviews: Biochemistry, 6th edition - Ch.1 PDF pp.10-29; Ch.2 pp.30-54

Qualitative tests: discriminate the molecular target

SOURCE BASIS: lecture

Test	Chemical target/principle	Positive result	Important limitation
Ninhydrin	Free alpha-amino groups; oxidative reaction and colored condensation product	Usually blue-violet; proline/hydroxyproline yellow	Other primary amino compounds/ammonia can react; not unique identification of one amino acid.
Xanthoproteic	Nitration of suitably reactive aromatic rings by concentrated nitric acid	Yellow; alkali intensifies to orange	Phe can be weak/negative under usual test conditions despite being aromatic.
Pauly	Diazotized sulphanilic acid couples with Tyr phenol or His imidazole in alkaline medium	Red azo color	Does not distinguish Tyr from His alone.
Hopkins-Cole	Trp indole reacts with glyoxylic reagent at concentrated sulfuric-acid interface	Violet/purple interface ring	A negative result does not mean the sample contains no protein; its Trp content matters.
Biuret	Cu ²⁺ complex with at least two peptide bonds in strong alkali	Violet	Free amino acids and a simple dipeptide do not give the classic positive protein response.
TCA precipitation	Strong acid reduces protein solubility and disrupts native structure	White precipitate/turbidity	Shows precipitable protein behavior, not one particular amino acid residue.

Use a pattern, not one tube

Tyrosine can be both xanthoproteic-positive and Pauly-positive; histidine is Pauly-positive without being a standard xanthoproteic target; tryptophan is Hopkins-Cole-positive. A protein is a mixture of residues, so its side-chain tests depend on composition and accessibility.

Sources for this section

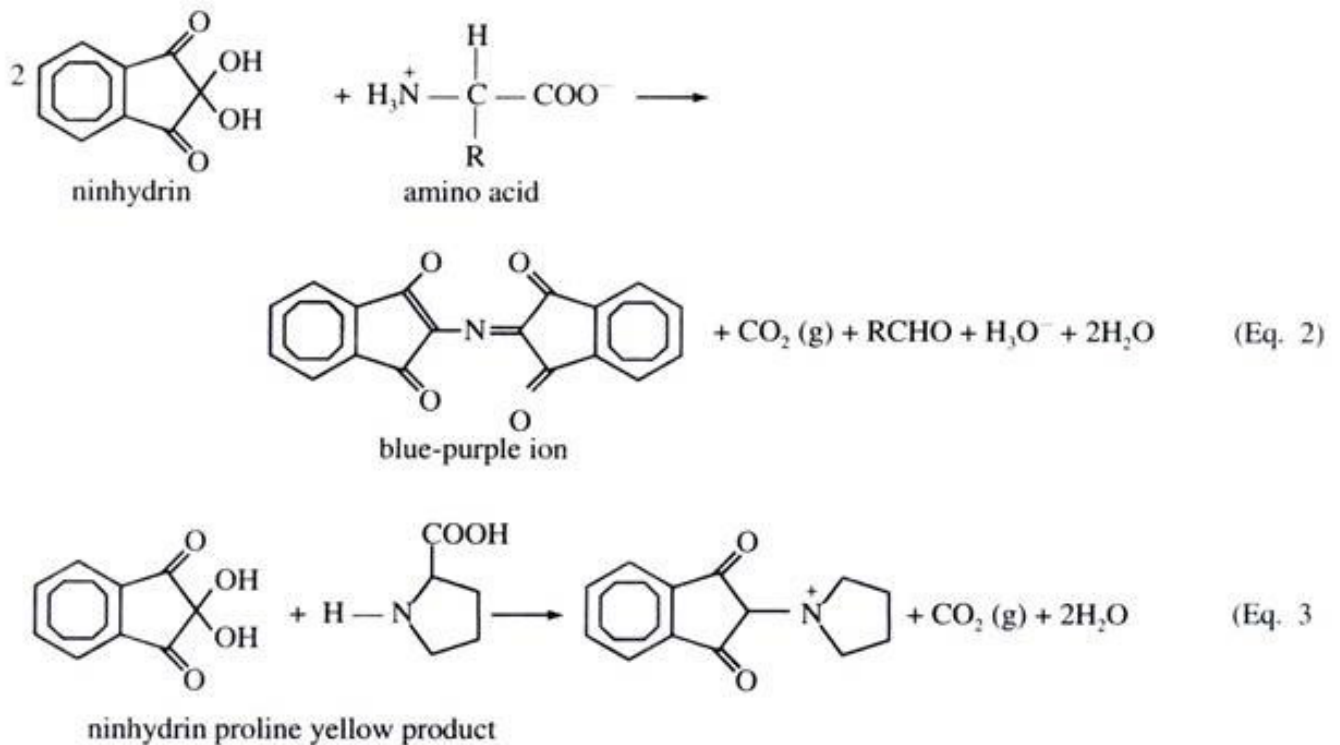
04A - Amino acid and protein qualitative tests - Ebrahimi.pptx - slides 1-29

04B - Amino acid and protein qualitative tests - Emamgholipour.pdf - PDF pp. 1-28

Ninhydrin: principle, exact protocol and interpretation

SOURCE BASIS: lecture

The usual alpha-amino-acid reaction involves oxidative deamination/decarboxylation and development of Ruhemann purple. The aldehyde derived from the amino acid has one fewer carbon than the starting amino acid. The reaction detects available amino groups rather than simply a peptide bond; free amino acids generally react strongly. Proline and hydroxyproline produce a yellow product because their secondary amino chemistry differs.



Original course reaction diagram: follow the amino-acid carbon skeleton and the colored product. The diagram supplies the chemistry; the two teacher versions differ in heating duration.

Identify: Ninhydrin; Alpha-amino acid; Carbon dioxide; Aldehyde; Ruhemann purple

Figure source: Emamgholipour amino-acid/protein qualitative tests, PDF p. 16, extracted reaction diagram

Component/step	Ebrahimi version	Emamgholipour version
Amino-acid solution	2 mL	2 mL
Ninhydrin	0.5 mL of 0.1% ninhydrin in acetone	0.5 mL of 0.1% ninhydrin in acetone
Heat in boiling-water bath	5 minutes	10 minutes
Observe	Blue-violet for usual alpha-amino acids; secondary-amino exceptions	Same principle

Protocol difference is real

Do not average the two heating times or claim one shared course protocol. Use the teacher/examiner-specified deck. Underheating, old reagent, contamination and inappropriate blank interpretation can alter the apparent result.

Check yourself: A sample is ninhydrin-positive but Biuret-negative. Must it be a protein?

No. Free amino acids can show exactly this pattern because they have reactive amino groups but lack the multiple peptide bonds required for the classic Biuret response.

Sources for this section

04A - Amino acid and protein qualitative tests - Ebrahimi.pptx - slides 16-17

04B - Amino acid and protein qualitative tests - Emamgholipour.pdf - PDF pp. 16-17

Xanthoproteic: aromatic nitration and the phenylalanine trap

SOURCE BASIS: lecture

Concentrated nitric acid nitrates sufficiently reactive aromatic rings. Yellow nitrated products develop; addition of alkali produces a stronger orange appearance. Tyrosine and tryptophan are the usual useful targets. The course groups phenylalanine with aromatic amino acids, which is structurally correct, but its unsubstituted phenyl ring is much less readily nitrated under ordinary qualitative-test conditions. It can therefore be weak or negative rather than a reliably strong positive.

Step	Shared / different protocol
Initial mixture	2 mL amino-acid solution + 2 mL concentrated nitric acid
Heating	Boiling-water bath for 5 minutes
Alkali stage	Emamgholipour explicitly says cool and add 2 mL 40% NaOH; Ebrahimi step list does not include this explicit confirmation
Positive interpretation	Yellow after nitration, orange after the specified alkaline stage

A negative test does not exclude all aromatic amino acids

Do not answer that phenylalanine is nonaromatic because it reacts poorly. The chemical structure and the sensitivity of this laboratory reaction are different questions. Strong acid/base steps must follow the supervised protocol, including cooling before the specified alkali addition.

Sources for this section

04A - Amino acid and protein qualitative tests - Ebrahimi.pptx - slides 18-19

04B - Amino acid and protein qualitative tests - Emamgholipour.pdf - PDF pp. 13-14

Mohanlal Sukhadia University: Xanthoproteic test - principle and phenylalanine limitation - https://mlsu.ac.in/econtents/1848_Xanthoproteic-test.pdf

Pauly test: tyrosine or histidine

SOURCE BASIS: lecture

Sulphanilic acid is diazotized using sodium nitrite in acidic conditions. The diazonium reagent then couples with the phenolic ring of tyrosine or the imidazole system of histidine in alkaline medium, producing a red azo dye. Alkalinity and reagent order are part of the reaction design. The table below follows Ebrahimi's first-two-component order; Emamgholipour places sulphanilic reagent in the tube before the amino-acid solution. Both then add nitrite, wait two minutes and add carbonate. Preserve the examiner's specified version.

Order	Course mixture/action
1	1 mL amino-acid solution
2	Add 1 mL sulphanilic-acid reagent
3	Add 1 mL 5% sodium nitrite
4	Vortex and wait 2 minutes
5	Add 1 mL 10% sodium carbonate; observe red color

Check yourself: An unknown amino acid is Pauly-positive. Can you call it tyrosine immediately?

No, histidine also gives a positive reaction. Use additional independent chemistry, such as a suitable aromatic-residue test and known controls, instead of treating one overlapping reaction as unique identification.

Sources for this section

04A - Amino acid and protein qualitative tests - Ebrahimi.pptx - slides 20-21

04B - Amino acid and protein qualitative tests - Emamgholipour.pdf - PDF pp. 18-19

Hopkins-Cole: the tryptophan interface ring

SOURCE BASIS: lecture

The indole group of tryptophan reacts with glyoxylic acid in the presence of concentrated sulfuric acid to form a violet product at the interface. The course identifies its Hopkins-Cole reagent as magnesium glyoxylate. The visible interface is important to the test: a mixed bulk color and a sharply observed junction are not the same observation.

Step	Course procedure
Prepare	Mix 1 mL amino-acid solution with 1 mL Hopkins-Cole reagent
Layer	Carefully introduce 2 mL concentrated sulfuric acid down the wall of the inclined tube under supervision
Observe	Violet/purple ring at the liquid interface

Step	Course procedure
Avoid	Do not vigorously shake the concentrated-acid interface or direct the tube toward anyone



Source photograph of the purple interface reaction. Use the layer junction as the observation site; do not infer a quantified concentration from color depth.

Identify: Acid interface; Purple reaction zone; Comparison tube

Figure source: Emamgholipour amino-acid/protein qualitative tests, PDF p. 20, embedded photograph

Check yourself: Why might a protein be Biuret-positive but Hopkins-Cole-negative?

Biuret responds to its peptide bonds. Hopkins-Cole requires tryptophan indole groups, so a protein with little or no tryptophan, or a technically failed reaction, may not show the ring. The tests ask different chemical questions.

Sources for this section

04A - Amino acid and protein qualitative tests - Ebrahimi.pptx - slides 22-23

04B - Amino acid and protein qualitative tests - Emamgholipour.pdf - PDF pp. 20-21

Qualitative Biuret and TCA precipitation

SOURCE BASIS: lecture

Test	Course mixture	Expected observation
Qualitative Biuret	2 mL protein solution + 2 mL 10% NaOH; slowly add 1 mL 1% CuSO ₄	Violet peptide-copper complex compared with reagent control
TCA, Ebrahimi	1 mL protein solution + 1 mL 15% TCA	White protein precipitate
TCA, Emamgholipour	2 mL protein solution + 2 mL 15% TCA	White protein precipitate

The two TCA versions retain the same 1:1 sample-to-reagent ratio but different absolute volumes. Do not silently merge them. TCA precipitation separates protein from a soluble supernatant under the specified conditions; it is not a residue-specific color test. Denaturation can expose or rearrange groups and disrupt solubility, but is distinct from complete hydrolysis into amino acids. Both qualitative Biuret decks describe a blue/purple junction ring. The chemical positive is formation of the peptide-copper complex; unlike the Hopkins-Cole acid-interface test, a ring is not the defining molecular requirement of every Biuret method. Report the observation required by the particular teaching procedure and distinguish it from reagent blue alone.

Qualitative versus quantitative Biuret

The qualitative protocol uses separate NaOH and CuSO₄ volumes. The spectrophotometric total-protein assay uses a prepared Biuret reagent, matched blank/standard, a 15-minute incubation and 540 nm measurement. Do not exchange their mixtures while retaining the quantitative formula.

Check yourself: What pattern would you expect for a simple free amino-acid mixture without peptide contamination?

Often ninhydrin-positive, with residue-specific tests depending on composition, but no classic protein Biuret response and no ordinary protein precipitation pattern with TCA. Unexpected turbidity or color requires a control and a check of contamination/reagents.

Sources for this section

04A - Amino acid and protein qualitative tests - Ebrahimi.pptx - slides 26-29

04B - Amino acid and protein qualitative tests - Emamgholipour.pdf - PDF pp. 24-27

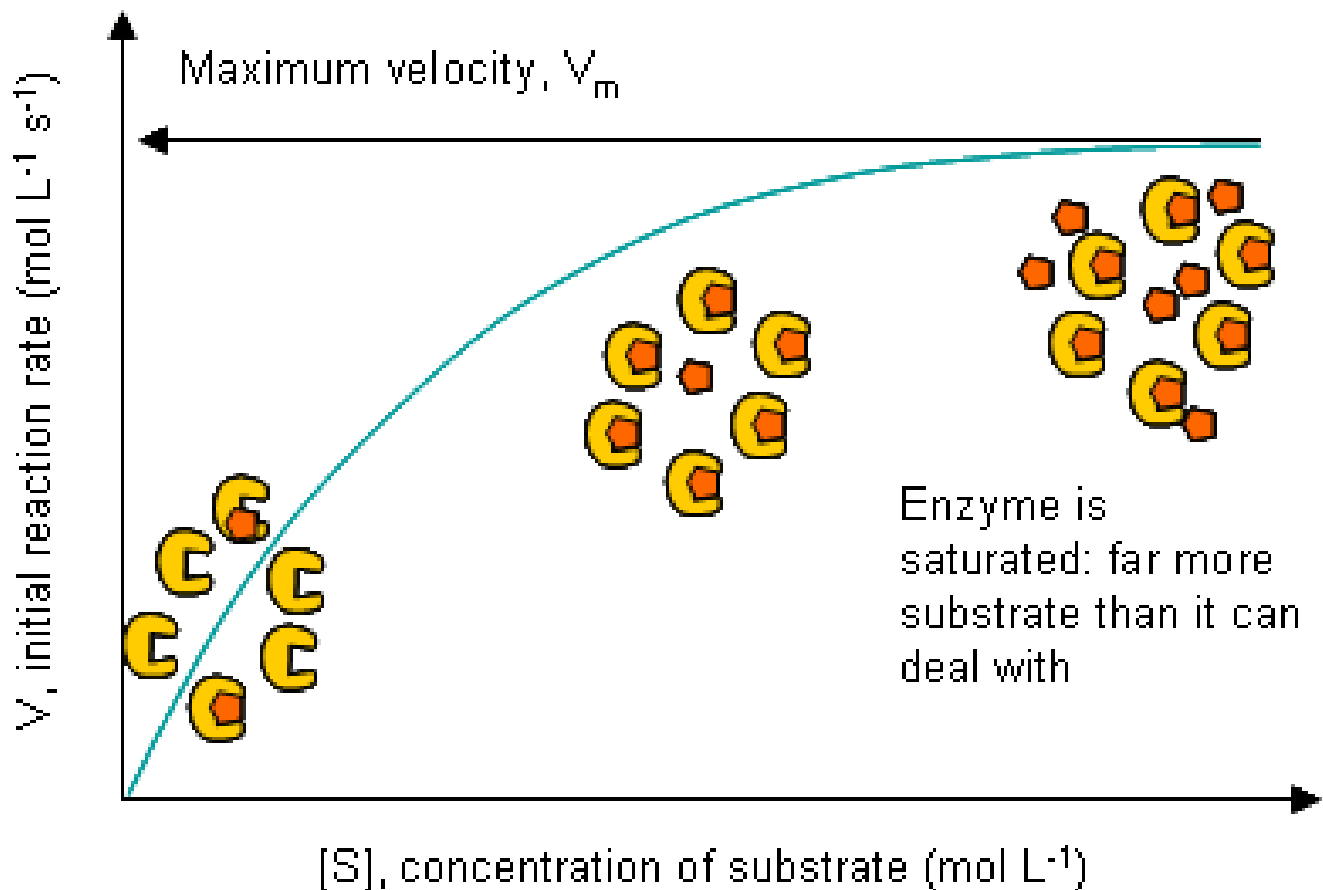
Enzyme activity: specificity, temperature, pH and amount

SOURCE BASIS: lecture + book

An enzyme speeds approach to equilibrium by lowering activation free energy; it does not change the reaction ΔG or equilibrium position. Most enzymes are proteins, while catalytic RNAs are exceptions. An active site binds substrate and supports catalysis through its geometry and chemical groups. Induced fit emphasizes conformational adjustment rather than an absolutely rigid lock.

An apoenzyme lacks a required nonprotein component; the active holoenzyme includes it. Cofactor is broadly used for a required nonprotein helper, including metal ions or organic coenzymes; the book uses the narrower metal/coenzyme distinction. A tightly bound prosthetic group differs from a diffusible cosubstrate. Enzyme amount, substrate amount and reaction conditions are separate variables.

Variable	Expected principle	Do not overgeneralize
More active enzyme	Higher initial rate and V_{max} if substrate is not limiting	A fixed-volume clot endpoint is not a direct assay of every catalytic event.
More substrate	Rate rises toward saturation for a Michaelis-Menten enzyme	At saturation more substrate cannot indefinitely increase rate.
Temperature rise	Can accelerate reaction before instability/denaturation dominates	There is no universal single temperature at which all enzymes instantly die.
pH change	Changes ionization of enzyme/substrate and potentially protein structure	The optimum is enzyme- and assay-specific; pepsin is not optimized at neutral pH.
Inhibitor or lost cofactor	Can reduce observed activity	A low rate alone does not identify the inhibition mechanism.



Original course saturation graph: the plateau reflects a finite enzyme catalytic capacity, not disappearance of substrate from the graph.

Identify: Initial reaction rate; Substrate concentration; V_{max} ; Enzyme saturation

Figure source: Enzyme practical, slide 8, embedded graph

Sources for this section

05 - Enzyme practical.pptx - slides 1-15

Lippincott Illustrated Reviews: Biochemistry, 6th edition - Ch.5 PDF pp.105-115

Kinetics and inhibition: a rigorous extension for hard questions

SOURCE BASIS: lecture + book

For the simple Michaelis-Menten model, $v_0 = V_{max}[S]/(K_m + [S])$, with initial velocity, substrate in excess of enzyme and a quasi-steady enzyme-substrate complex. At $[S] = K_m$, v_0 is half V_{max} . At $[S]$ much below K_m , rate is approximately proportional to substrate; far above K_m , rate approaches V_{max} and is nearly independent of substrate. Doubling active enzyme doubles V_{max} under otherwise comparable conditions but does not itself change K_m . K_m is often described as an affinity-related quantity, but mathematically includes both dissociation and catalytic rate constants and is not universally identical to the binding dissociation constant.

Inhibitor pattern	Apparent K_m	Apparent V_{max}	High substrate
Competitive	Increases	Unchanged	Can overcome the ideal reversible inhibition

Inhibitor pattern	Apparent K_m	Apparent V_{max}	High substrate
Pure noncompetitive	Unchanged	Decreases	Cannot restore original V_{max}
Uncompetitive: broader kinetic distinction	Decreases	Decreases by the same factor	Binds ES; not the same mechanism as competition for free E
Mixed: broader kinetic distinction	May increase or decrease	Decreases	Unequal binding to free E and ES; pure noncompetitive is the equal-affinity special case

The double-reciprocal relation is $1/v_0 = (K_m/V_{max})(1/[S]) + 1/V_{max}$. Its y-intercept is $1/V_{max}$, x-intercept is $-1/K_m$ and slope is K_m/V_{max} . Competitive lines meet at the same y-intercept; pure noncompetitive lines share the same x-intercept. These ideal plots are teaching tools; transformed experimental errors can make quantitative fitting less reliable than direct model fitting.

Different binding site does not prove pure noncompetitive kinetics

Allosteric regulation and inhibition patterns are not synonyms. An effector at another site may change affinity, catalytic capacity or both. The course urease tube experiment does not measure multiple substrate concentrations and therefore cannot establish a K_m/V_{max} inhibition class.

Check yourself: V_{max} is 100 units and K_m is 2 mmol/L. What rate is expected at substrate 6 mmol/L?

$v_0 = 100 \times 6/(2+6) = 75$ units. Raising substrate from 2 to 6 mmol/L triples substrate but changes rate from 50 to 75 units, not to 150, because the enzyme is approaching saturation.

Sources for this section

05 - Enzyme practical.pptx - slides 6-14

Lippincott Illustrated Reviews: Biochemistry, 6th edition - Ch.5 PDF pp.117-127

Milk-clotting experiment: temperature

SOURCE BASIS: lecture

Rennin is not renin

The course milk-clotting enzyme is rennin/chymosin. Renin is the renal enzyme of the renin-angiotensin system. The slide spelling should not make you transfer the milk reaction to kidney physiology.

Chymosin cleavage destabilizes milk casein micelles and permits clot formation. The measured endpoint is visible clotting, which includes physical aggregation after the enzymatic step. Keep milk composition, volumes, mixing and endpoint criteria comparable. The course protocol pre-incubates enzyme and milk separately, then combines matched tubes.

Condition	A tube	B tube	Procedure
37 C	0.5 mL rennin	2 mL milk	Pre-incubate separately 2 minutes, mix, return to matched bath and time clotting
80 C	0.5 mL rennin	2 mL milk	Same sequence at 80 C
Room temperature	0.5 mL rennin	2 mL milk	Same sequence at room temperature

A reasonable expectation is faster clotting near the suitable warm condition than at room temperature, while substantial prior enzyme heating at 80 C can impair activity. But actual clot times must be measured, not fabricated; heat also changes milk proteins and the clot endpoint. The high-temperature tube tests exposure plus subsequent reaction conditions, not merely an abstract temperature coefficient.

Check yourself: Why does preheating enzyme and milk separately matter?

If they were mixed before the defined pre-incubation, catalysis could occur during temperature equilibration. Separate incubation better defines when the reaction starts and what temperature each component had on mixing.

Sources for this section

05 - Enzyme practical.pptx - slides 15-16

Milk-clotting experiments: enzyme versus substrate concentration

SOURCE BASIS: lecture

Enzyme-concentration condition	Milk in A (mL)	Rennin in B (mL)	Water in B (mL)
Highest enzyme	2	1.00	0
Half enzyme	2	0.50	0.50
Quarter enzyme	2	0.25	0.75

For the enzyme-concentration comparison, every final mixture is 3 mL and contains the same milk volume. Pre-incubate matched tubes at 37 C for 2 minutes, combine and measure clot time. The relative amounts of enzyme are 1, 0.5 and 0.25. Higher active enzyme is expected to reach a comparable clot endpoint sooner. The reciprocal of clot time is only an approximate comparative activity proxy when endpoint and other conditions are comparable; it is not an automatically calibrated amount of product per minute.

Substrate-concentration condition	Milk in A (mL)	Water in A (mL)	Rennin in B (mL)
Highest milk	2.00	0	1

Substrate-concentration condition	Milk in A (mL)	Water in A (mL)	Rennin in B (mL)
Half milk	1.00	1.00	1
Quarter milk	0.50	1.50	1

The substrate comparison again keeps final volume 3 mL and enzyme amount constant, with the same 37 C separate pre-incubation for 2 minutes. The theoretical initial catalytic rate can rise with substrate toward saturation. However, milk dilution also changes casein aggregation and visibility of a clot. Do not claim that a clot-time ranking alone provides a clean Michaelis-Menten curve or a numerical K_m .

Check yourself: Why add water to the lower-rennin tubes rather than simply use smaller total volumes?

It holds final reaction volume and milk concentration constant while changing enzyme amount. Otherwise concentration and mixing/endpoint changes would confound the intended independent variable.

Sources for this section

05 - Enzyme practical.pptx - slides 17-18

Urease: substrate specificity, inhibition and an indicator endpoint

SOURCE BASIS: lecture

Urease hydrolyzes urea to carbon dioxide and two ammonia molecules. Ammonia reacts with water and raises pH; phenol red reports this alkalinization by becoming red/pink relative to the starting control. The indicator is not a direct spectroscopic measurement of enzyme concentration. Thiourea is a structurally related comparison compound but does not serve as the course urease substrate.

Tube	Substrate/comparison	Additional reagent	Urease	After mixing
1: reaction	2 mL urea	2 mL water	2 mL	Mix 30 seconds; add 0.5 mL phenol red
2: inhibitor	2 mL urea	2 mL zinc sulfate	2 mL	Same sequence
3: specificity	2 mL thiourea	2 mL water	2 mL	Same sequence

Comparison	Expected teaching interpretation	Caveat
Tube 1	Ammonia generation raises pH and changes indicator color	Check starting pH and a matched baseline.

Comparison	Expected teaching interpretation	Caveat
Tube 2 versus 1	Zinc sulfate reduces observed urease activity, so less alkaline response is expected	The experiment alone does not establish competitive versus noncompetitive kinetics; reagent pH may also influence indicator.
Tube 3 versus 1	Lack of the urea-like response supports substrate specificity	No color change is only interpretable if enzyme and indicator controls work.

Controls make the causal claim

For rigorous inference, matched enzyme-free and reagent/pH controls distinguish chemical background from enzyme-generated alkalinity. The slide lists three teaching tubes, not a complete validated clinical assay. Keep observed colors separate from an inferred molecular mechanism.

Check yourself: The zinc tube remains yellow. Does this prove zinc occupies the urea-binding site?

No. It shows reduced alkaline response under those conditions. Binding-site competition would require appropriate kinetic evidence across substrate concentrations and exclusion of pH/color interference.

Sources for this section

05 - Enzyme practical.pptx - slides 19-21

Flame photometry: emission, instrument and wavelength

SOURCE BASIS: lecture

Flame photometry measures light emitted by excited atoms. A solution is aspirated and nebulized, solvent evaporates, salts are atomized and some atoms are excited by flame energy. On returning toward a lower energy state, they emit characteristic wavelengths. A wavelength selector and detector measure the relevant intensity. Within a validated range and controlled flame/sample conditions, emission intensity relates to concentration. This is not Beer-Lambert absorbance and does not use a transmitted-beam ratio in the same way.

Element	Course analytical wavelength	Course flame-color description
Sodium	589 nm	Yellow
Potassium	766 nm	Violet/lilac
Lithium	670 nm	Red
Calcium	622 nm	Orange
Barium	554 nm	Lime green

Potassium color and analytical wavelength are different facts

The strong potassium analytical line around 766 nm is near infrared/deep red, not violet light. A perceived lilac flame comes from the overall visible emissions. Do not use the table to assert that 766 nm is a violet wavelength.

The calcium channel measures a band, not the main atomic line

The course 622 nm calcium value is preserved. The Jenway manufacturer manual specifies that this is calcium hydroxide band emission; the principal calcium atomic emission is around 423 nm. Thus a simple flame-photometer teaching table can contain both atomic lines and a molecular-band measurement.

- Learn the functional chain: sample uptake/nebulizer, burner/flame, wavelength selection, detector, signal processing/readout. A blockage, aspiration change or unstable gas supply changes signal even when sample concentration is unchanged.
- A reagent blank establishes background; known standards establish response; appropriate quality-control material tests ongoing performance. Use the instrument-specific SOP for standard concentrations, dilution and acceptance limits, which are not supplied here.
- Interferences include spectral overlap, matrix effects on aspiration, ionization and chemical effects. The method is useful for selected easily excited elements; do not infer that every metal is measured equally well by a simple flame photometer.
- Basic clinical flame methods commonly emphasize Na and K, with Li and selected other analytes depending on the instrument. The slide broad statement about magnesium is not sufficient evidence of a routine magnesium flame protocol.

Sources for this section

06 - Flame photometry and body electrolytes.pptx - slides 1-3, 6-9; slide 6 is hidden in the original deck

Jenway PFP7/PFP7C manufacturer instruction manual - PDF p.20, section1.3: calcium hydroxide band at 622nm - https://www.keison.co.uk/products/jenway/PFP7_Manual.pdf

Electrolytes: biological roles and analytical pitfalls

SOURCE BASIS: lecture

Sodium is the main extracellular cation and strongly relates to extracellular osmolality and water distribution. Potassium is predominantly intracellular and is critical for membrane potential and excitability. A serum concentration is influenced by water balance, shifts between compartments, intake/losses, kidney handling and sample quality; it is not a direct measurement of total body stores. The slide association of hypernatremia simply with edema is misleading: hypernatremia commonly reflects water deficit relative to sodium, though clinical volume states vary.

Routine contemporary clinical Na/K analysis commonly uses ion-selective electrodes. Direct ISE analyzes undiluted material; indirect ISE uses dilution. Excess lipid or protein can produce pseudohyponatremia with methods that assume a normal plasma-water fraction, including indirect ISE, whereas direct ISE avoids that particular dilution/water-fraction artifact. This does not mean that every ISE measurement is universally less precise than flame photometry, as the slide suggests. Accuracy and precision depend on the system and its quality control.

Observation/problem	Reasoning
Unexpected high potassium in a damaged sample	Hemolysis can release intracellular potassium; verify sample quality before assuming a biological disorder.
Low sodium with marked hyperlipidemia/proteinemia	Consider method-dependent pseudohyponatremia and compare with an appropriate direct method; do not diagnose from this clue alone.
Different units	For monovalent Na ⁺ and K ⁺ , mmol/L equals mEq/L numerically. For Ca ²⁺ , 1 mmol/L corresponds to 2 mEq/L.
Serial sample dilution	Apply the validated dilution factor to an in-range instrument result; do not extrapolate from an overloaded signal.
Known standard reads incorrectly	Check calibration, blank, aspiration, contamination and instrument stability before interpreting an unknown.

Check yourself: Why is 0.9% sodium chloride approximately 154 mmol/L NaCl?

0.9% w/v is 0.9 g per 100 mL, or 9 g/L. Divide by NaCl molar mass about 58.44 g/mol: 0.154 mol/L. Dissociation supplies approximately 154 mmol/L Na⁺ and 154 mmol/L Cl⁻. This is a unit calculation, not a prescription for preparing or administering an infusion.

Sources for this section

06 - Flame photometry and body electrolytes.pptx - slides 4-5, 8-11

IFCC recommendation on sodium and potassium by ion-selective electrodes in undiluted serum, plasma or whole blood - 2000 recommendation; activity, calibration and reporting - <https://pubmed.ncbi.nlm.nih.gov/11140625/>

CLSI C29: Standardization of sodium and potassium ion-selective electrode systems - method standardization and reference framework - <https://clsi.org/shop/standards/c29/>

Integrated difficult practical checks

SOURCE BASIS: lecture + book

If asked...	A complete answer includes...
Why choose lambda max?	Higher analytical sensitivity for that chromophore, fixed conditions across standards/unknowns, and actual measured maximum rather than guessed visible color.
What must be changed when an unknown exceeds calibration?	Perform an appropriate documented dilution, remeasure inside the validated range and apply the factor.
Why a protein can test differently in different reactions?	Peptide backbone versus residue-specific chemistry, composition, denaturation/solubility and technical controls.
Why enzyme activity changes with pH?	Ionization of catalytic/substrate groups plus possible structural destabilization, not simply more or fewer hydrogen ions as an unexplained slogan.
Can a clot time identify K_m ?	Not from this setup alone: the endpoint involves physical aggregation, and rigorous initial-rate substrate series are missing.
What does an inhibitor tube prove?	An altered measured outcome under specified conditions; mechanism needs stronger controls and kinetic evidence.
What is the central flame-versus-absorbance distinction?	Excited-atom emitted intensity versus logarithmic attenuation of an incident beam.
Which course numbers must not be silently corrected into a new protocol?	Ninhydrin 5 versus 10 minutes; xanthoproteic alkali step; TCA 1+1 versus 2+2 mL. Preserve the relevant teacher version.

Check yourself: A student obtains high total protein, a cloudy sample and an out-of-range absorbance. What should the report avoid?

Avoid immediately calling it dehydration or myeloma. Turbidity can alter optical measurement and the concentration is outside validated calibration. The analytical issue must be resolved under the assay SOP, with appropriate dilution/interference handling and controls, before clinical interpretation.

Check yourself: A student explains that an enzyme makes an energetically unfavorable overall reaction favorable. Correct this precisely.

The enzyme lowers the activation barrier and speeds both directions toward the existing equilibrium. It does not change the overall reaction free energy. An unfavorable reaction can proceed by coupling to a favorable process or by cellular concentration conditions, which are separate from catalysis itself.

Sources for this section

01 - Laboratory safety and equipment.pdf - topics reviewed above

02 - Titration practical and calculations.pdf - topics reviewed above

03 - Spectrophotometry and Beer-Lambert law.pptx - topics reviewed above

04A - Amino acid and protein qualitative tests - Ebrahimi.pptx - topics reviewed above

04B - Amino acid and protein qualitative tests - Emamgholipour.pdf - topics reviewed above

05 - Enzyme practical.pptx - topics reviewed above

06 - Flame photometry and body electrolytes.pptx - topics reviewed above

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Coverage limits and next sources

- No practical lecture recordings/transcripts are present in this folder.
- The laboratory equipment and titration items are links to earlier BIOCHEM practical files, retained as foundations rather than proof they are newly examinable.
- Ninhydrin heating time, the explicit alkali step in xanthoproteic testing and TCA volumes differ between the two supplied teacher versions. Follow the examiner-specified version; differences are documented below.
- No instrument-specific flame-photometer settings, patient-sample dilution, calibration concentrations or quality-control acceptance limits are supplied. These must not be invented.
- Slides show predicted or illustrative results but do not contain your measured laboratory observations. Numerical examples below are teaching calculations, not recorded experiment results.

Local source inventory

Exact paths and source hashes are stored in `_build/source-manifest.json`. Section references distinguish original evidence from book additions.

Biochemistry [Lippincott's Illustrated Reviews Series] Denise R. Ferrier PhD.pdf

01 - Laboratory safety and equipment.pdf

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